A voltage source $\underline{E} = 7 + \frac{14\sqrt{3}}{2} j$[....1.] powers a circuit, determining a current flow through this one of value $\underline{I} = 2e^{j\frac{\pi}{4}}$[....2.]. Compute the complex power of this circuit, $\underline{S}$ 3, and explain the significance of the real part 4 and the imaginary part of this power 5, and state their measurements units 6. Find the sinusoidal exression of the voltage source e(t) 7, and compute its value when t=7[ms] 8 if the circuit works on the frequency f=50[...9..]. One circuit element is a resistance. Compute its value starting from the active power 10. State the measurement units for the following quatities: $\overline{H}$ *and* ε 11 and 12.

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1. Volt, V
2. Ampere, A
3. $\underline{S} = \underline{E} \cdot \underline{I}^* = (7 + \frac{14\sqrt{3}}{2}j) \cdot 2e^{-j\frac{\pi}{4}} = 14e^{j\frac{\pi}{3}} \cdot 2e^{-j\frac{\pi}{4}} = 28e^{j\left(\frac{\pi}{3}-\frac{\pi}{4}\right)} = 28e^{j\frac{\pi}{12}} = 27{,}04 + 7{,}24j$
4. P – active power
5. Q – reactive power
6. W (Watt) and VAr (Volt-Ampere reactive)
7. $e(t) = 14\sqrt{2}\sin\left(\omega t + \frac{\pi}{3}\right)$
8. $e(7ms) = 14\sqrt{2}\sin\left(2\pi \cdot 50 \cdot 7 \cdot 10^{-3} + \frac{\pi}{3}\right) \cong 19{,}79\sin(\pi \cdot 1{,}03) \cong [A]$
9. Hz - Hertz
10. $P = R \cdot I^2 = 27{,}04 \rightarrow \quad R = \frac{P}{I^2} = \frac{27{,}04}{2^2} = \frac{27{,}04}{4} = 6{,}76[ohm]$
11. A/m
12. 12. F/m